US corn yield in August: a scored 80% band

Desk memo. Public USDA data only, rebuilt as it stood on each release morning. Every number below is generated by the repository's `make memo`.

Question

The desk prices off the August USDA corn yield print. To gauge potential deviation, a band is constructed around this yield, using the 10th and 90th percentiles of USDA's past August misses against the January final. This 80% band is applied to the August yield. The test assesses whether the January final yield falls within this band, based on 22 past seasons from 2004 to 2025.

August 2025 answer

Object	Unit	USDA point	q10	q50	q90	80% band	Truth (Jan first-final)	Inside
M1	bu/ac	188.8	178.6	188.8	200.6	[178.6, 200.6]	186.5	yes
PROD	Mbu	16742	15592	16768	17893	[15592, 17893]	17021	yes

On August 12, 2025, the USDA estimated corn yield at 188.8 bu/ac, with our 80% band ranging from 178.6 to 200.6 bu/ac. The corresponding production estimate was 16,742 Mbu, with our band at 15,592 to 17,893 Mbu. By January, the final yield was 186.5 bu/ac and production was 17,021 Mbu, both within their respective bands. This single season's result is merely a check on the arithmetic, with the true test of our band's performance lying in its 22-season track record.

Stocks-to-use decision line

The decision line for 2025/26 US corn stocks-to-use is 13.3% according to the USDA, while our q10 estimate is 6.1%. This estimate is based on production quantiles passing through the USDA's August balance sheet, with beginning stocks, imports and total use held fixed, so ending stocks move one for one with production. Our calculation shows that the downside yield tail halves stocks-to-use in 2025. This is notable because, as Wright (2011) notes, price becomes steep at low stocks, making the thin tail important. The q50 is 13.4% and the q90 is 20.5%.

How the band scored, 2004–2025

Model	Horizon	Level	n	x	Coverage	CP90	Wording
M1	May	80%	22	20	90.9%	[74.1%, 98.4%]	not distinguishable
M1	Jun	80%	22	18	81.8%	[63.1%, 93.5%]	not distinguishable
M1	Jul	80%	22	18	81.8%	[63.1%, 93.5%]	not distinguishable
M1 (primary)	Aug	80%	22	21	95.5%	[80.2%, 99.8%]	too wide
M1	Sep	80%	22	21	95.5%	[80.2%, 99.8%]	too wide
M1	Oct	80%	20	17	85.0%	[65.6%, 95.8%]	not distinguishable
M1	Nov	80%	22	17	77.3%	[58.0%, 90.6%]	not distinguishable
PROD	Aug	80%	22	20	90.9%	[74.1%, 98.4%]	not distinguishable
B1	Aug	90%	22	21	95.5%	[80.2%, 99.8%]	descriptive

The primary pre-registered result shows that 21 out of 22 seasons fell within the August band, resulting in a 95.5% coverage rate. The 90% Clopper-Pearson interval for this coverage rate is 80.2%–99.8%, with the pre-registered

wording describing the band as “too wide”. The estimator used is conservative by construction, so at the nominal 80% rate it is expected to cover 84.2% of seasons at these sample sizes; that accounts for about 4 points of the excess coverage. A comparison between M3 and M1 in August found no detectable difference, with a one-sided p-value of 0.888. Out of 33 exploratory tests, none had Holm-adjusted p-values below 0.05. Production is scored at 80% coverage, while its 50% coverage and the yield band’s 50% coverage are reported descriptively and with exploratory wording, respectively.

When this fails

The band breaks in two failure modes: drought springs and late surprises. In 2012, a drought year, May, June, and July origins were missed, with final yields beyond the 50% and 80% bands. However, August origins were covered. Late-season surprises also fall outside the bands, as seen in November 2020, which sat outside the 80% band. The stocks-to-use line is flawed, ignoring demand response and breaking in 2011/2012-type years. Eleven WASDE and nine NASS releases survive only as corrected copies, with the originals not archived; they are flagged, and no USDA error is claimed. The band’s precision is noted, with a 90% Clopper-Pearson interval spanning 80.2% to 99.8% over 22 seasons. Stress years, such as 2012, 2019, and 2020, have been descriptively scored.

Port to other origins

The code is reused as is, with nothing Russia-specific, on Russian wheat production; the line shows the harness runs on a second series and makes no claim about effectiveness. Analyzing 18 seasons, from 2006 to 2023, the August USDA print’s distance to the settled figure shows an 80% band covering 18 out of 18 seasons, with a 90% Clopper-Pearson interval of 84.7% to 100% coverage. The verdict on the band is “too wide”. The estimator alone expects 84.4% coverage at 18 seasons, so about 4 points of the excess coverage come from it. This exploratory analysis can be applied to other origins, such as EU rapeseed or Brazil safrinha, using a desk’s own data.